

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Geology and Geophysics

2. **Subject Name:** Geophysical Signal Processing

L-T-P: 3-1-0

Credits:4

3. **Pre-requisites:** None

4. **Syllabus and reference books:**

Syllabus: Signals, Systems and Signal Processing: Definition & Classifications, Continuous-Time versus Discrete-Time Signals. Multichannel Multidimensional Signals. Analog-to-Digital and Digital-to-Analog Signal Conversions, Aliasing, Sampling Theorem. Frequency Analysis of Signals and Systems: Fourier series and Periodic Waveforms. Fourier Transform (CTFT), its Theorems, properties and applications. Convolution, Crosscorrelation, Autocorrelation. Power Density Spectrum, Power Spectral Density, Energy Spectrum, Parseval's Theorem. Discrete Fourier Transform (DFT), its Theorems, properties and applications, Circular Convolution. Fast Fourier Transform (FFT). STFT. Hilbert Transform and its properties. Hilbert-Huang Transform (Empirical Mode Decomposition) and its properties. Wavelet Transform (CWT & DWT) and its properties. Z-transform, its Theorems and properties, inverse Z-transform, Rational Z-transforms, Poles-Zeros. Principles of Digital Filters, Moving Average and Recursive filters. Windowing functions for filter design. Gibb's Phenomenon. Pole-zero technique for designing ARMA filters. FIR Systems and design of FIR Filters. IIR Systems and design of IIR Filters. Low-pass, High-pass and Bandpass Filters. Shaping and spiking filters using Wiener's principle of least square estimation. Deconvolution of Seismic data: Feed-forward and Feed-backward Systems, Spectral factorization and Frequency-domain Deconvolution, Signature Deconvolution, Time-variant Deconvolution, Predictive or gapped Deconvolution, Weiner-Hoff design of Deconvolution operators, Surface-consistent Deconvolution. Time-variant Spectral Whitening (TVSW).

Space for entering the syllabus

Reference Books:

- 1) Digital Signal Processing by John G. Proakis and Dimitris G. Manolakis
- 2) Applied Digital Signal Processing: Theory and Practice by Dimitris G. Manolakis and Vinay K. Ingle
- 3) Understanding Digital Signal Processing by Richard G. Lyons
- 4) Geophysical Signal Analysis by Enders A. Robinson and Sven Trietel

5. **Lecture-wise break-up:**

Sl. No.	Topic	No. of lectures
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1.	Signals, Systems and Signal Processing: Definition & Classifications, Continuous-Time versus Discrete-Time Signals. Multichannel Multidimensional Signals.	04
2.	Analog-to-Digital and Digital-to-Analog Signal Conversions, Aliasing, Sampling Theorem.	04
3.	Frequency Analysis of Signals and Systems: Fourier series and Periodic Waveforms. Fourier Transform (CTFT), its Theorems, properties and applications. Convolution, Crosscorrelation, Autocorrelation.	04
4.	Power Density Spectrum, Power Spectral Density, Energy Spectrum, Parseval's Theorem.	04
5.	Discrete Fourier Transform (DFT), its Theorems, properties and applications, Circular Convolution. Fast Fourier Transform (FFT).	04
6.	STFT. Hilbert Transform and its properties. Hilbert-Huang Transform (Empirical Mode Decomposition) and its properties. Wavelet Transform (CWT & DWT) and its properties.	04
7.	Z-transform, its Theorems and properties, inverse Z-transform, Rational Z-transforms, Poles-Zeros.	04
8.	Principles of Digital Filters, Moving Average and Recursive filters.	04
9.	Windowing functions for filter design. Gibb's Phenomenon. Pole- zero technique for designing ARMA filters.	04
10.	FIR Systems and design of FIR Filters. IIR Systems and design of IIR Filters. Low-pass, High-pass and Bandpass Filters.	04
11.	Shaping and spiking filters using Weiner's principle of least square estimation.	02
12.	Deconvolution of Seismic data: Feed-forward and Feed-backward Systems, Spectral factorization and Frequency-domain Deconvolution, Signature Deconvolution, Time-variant Deconvolution, Predictive or gapped Deconvolution, Weiner-Hoff design of Deconvolution operators, Surface-consistent Deconvolution. Time-variant Spectral Whitening (TVSW).	06
Total number of hours		48